DEPARTMENT OF BIOMEDICAL ENGINEERING

OBT 356 LIFESTYLE DISEASES

DAILY TEST – 3 QUESTION PAPER

PART A

(5×2 = 10)

YEAR/SEM: IV/VII

1. After an abnormal ECG, Mr. Sharma, a 60-year-old from Bengaluru, undergoes an echocardiogram. Explain how this test provides crucial insights into diagnosing and assessing the impact of coronary artery disease on his heart.
2. Mrs. Rani, a 55-year-old from Pune, is experiencing new onset discomfort. What are the two common symptoms that would immediately make her doctor consider Coronary Artery Disease (CAD) as a possible cause?
3. Mr. Gopal, a 62-year-old from Chennai recently diagnosed with CAD, is advised to make significant lifestyle changes. Describe two crucial lifestyle modifications you would recommend to him for managing his condition and improving his heart health.
4. Following his heart attack, Mr. Singh, a 68-year-old from Punjab, is advised to join a cardiac rehabilitation program. Explain what is cardiac rehabilitation and how it will benefit his recovery and long-term heart health.
5. Mr. Venkatesh, a 70-year-old from Hyderabad, has been prescribed medication for his Coronary Artery Disease (CAD). Describe two common classes of medications used to treat CAD and briefly explain their purpose in his treatment plan.

PART B

(2×15=30)

1. Ms.Priya, a 35-year-old professional from Bengaluru, has a family history of Coronary Artery Disease (CAD) and recent medical tests show that she has an unfavorable lipid profile, with elevated LDL cholesterol and reduced HDL levels. Her physician warns that, if unmanaged, these factors could increase her risk of atherosclerosis and early onset heart disease.Dietary habits play a crucial role in lipid metabolism and coronary artery health. Diets high in saturated fats and trans fats tend to increase LDL cholesterol, promoting plaque buildup in arteries, whereas unsaturated fats (mono- and polyunsaturated) help

improve lipid balance and protect cardiovascular function. For Ms. Priya, making dietary changes—such as reducing fried foods, processed snacks, and fatty meats while increasing nuts, seeds, fish, and olive oil—can be a powerful preventive measure. Evaluate how different types of dietary fats impact Ms. Priya's coronary artery health and justify how modifying her fat intake could significantly lower her risk of developing atherosclerosis.

2. Mr. Joseph, a 68-year-old man from Kochi, has been diagnosed with Coronary Artery Disease (CAD) after experiencing chest pain, breathlessness, and fatigue during routine activities. His doctors explain that CAD develops when plaques made of fat, cholesterol, and other substances build up inside the coronary arteries, restricting blood flow to the heart. They want Mr. Joseph to clearly understand the disease so that he can actively participate in his treatment and lifestyle planning.CAD is influenced by risk factors such as smoking, high cholesterol, hypertension, diabetes, obesity, and sedentary lifestyle. It often presents with angina, shortness of breath, palpitations, or even a heart attack in severe cases. Diagnosis is usually made through ECG, echocardiography, stress tests, coronary angiography, and blood tests. Treatment may include lifestyle modifications, medications, angioplasty, or bypass surgery. Preventive measures, such as regular exercise, balanced diet, quitting tobacco, and controlling cholesterol, are essential to reduce future risks. Create a comprehensive summary of Coronary Artery Disease in Mr. Joseph's case by defining the condition, organizing its causes and symptoms, outlining diagnostic techniques, and designing a concise plan that includes treatment options and preventive strategies.